

Phase $2\pi iq$ cancels the $1/(2\pi iq)$ from integrating H

Continuum identity

Why the gather kernel is g , not g'

phase $\partial_p \sim 2\pi iq$

spectral $\int = 1/(2\pi iq)$

product cancels $\Rightarrow$ no q

adjoint of scatter(g) = gather(g)

Backward: gather ψ with the atomic Gaussian

